

Cambridge (CIE) IGCSE Maths
Paper 3: Calculator (Set E)
Core

Friday 25 April 2025

Afternoon (Time: 1 hour 30 minutes)

Total marks

/ 80

Instructions

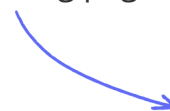
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

or visit the mock exams landing page for this course

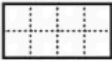
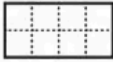
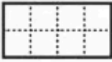
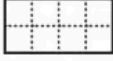
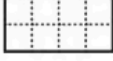


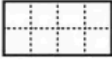
- 1 George, Louis and Beatriz have a café.

George records the number of each type of meal sold.

He draws a pictogram to show his results.

All rows are complete except for Salad.

Type of meal	Number of meals
Meat curry	 
Pasta	  
Vegetarian	
Salad	
Fish	 
Sandwich	

Key:  = 8 meals

- i) Six salads were sold.

Complete the pictogram.

[1]

- ii) Write down which type of meal was sold most.

[1]

- iii) Find the number of meals sold altogether.

[1]

(3 marks)

2 Write down

i) all the factors of 21,

[2]

ii) a prime number between 40 and 50.

[1]

(3 marks)

3 Write down the reciprocal of 10.

(1 mark)

4 Insert one pair of brackets into this statement to make it correct.

$$7 \times 5 - 2 + 3 = 42$$

(1 mark)

5 i) Write 4.2×10^{-3} as an ordinary number.

[1]

ii) Calculate $(8.1 \times 10^3) + (7.9 \times 10^4)$
Give your answer in standard form.

[2]

(3 marks)

- 6 The temperature at 0700 is -3°C .
This temperature is 11°C higher than the temperature at 0100.

Find the temperature at 0100.

..... $^{\circ}\text{C}$

(1 mark)

- 7 Thor changes 40 000 Icelandic Krona into dollars when the exchange rate is 1 krona = \$0.0099 .

Work out how many dollars he receives.

\$

(1 mark)

- 8 Simplify $8a + 3b - 2a + b$

(2 marks)

- 9 **Using a straight edge and compasses only**, construct the equilateral triangle ABC .
The base AB has been drawn for you.

A _____ B

(2 marks)

- 10 Find the value of $\sqrt{15}$.
Give your answer correct to 3 decimal places.

(2 marks)

11 (a) Dan either walks or cycles to school. The probability that he cycles to school is $\frac{1}{5}$.

Write down the probability that Dan walks to school.

(1 mark)

(b) There are 200 days in a school year.

Work out the expected number of days that Dan cycles to school in a school year.

(1 mark)

12 (a) A circle has diameter 6 cm.

Calculate the area of the circle.

Give the units of your answer.

(3 marks)

(b) Henry buys a circular mirror for the room.

The diameter of the mirror is 80 cm.

Calculate the circumference of the mirror.

..... cm

(2 marks)

13 Emma records the number of letters in each word in a sentence.

7	1	5	4	2	4	5	3	3	1	2	4
---	---	---	---	---	---	---	---	---	---	---	---

Find

i) the median,

[2]

ii) the mode,

[1]

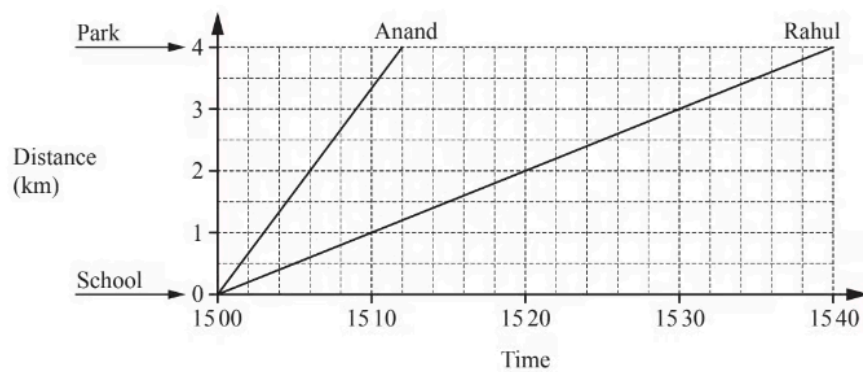
iii) the range.

[1]

(4 marks)

14 (a) Anand, Rahul and Samir go from school to the park each day.

One day, Anand cycles and Rahul walks. The travel graph shows their journeys.



i) Work out the speed that Anand cycles. Give your answer in kilometres per hour.

.....km/h [2]

ii) Find the number of minutes that Anand arrives at the park before Rahul.

..... min [1]

iii) Samir cycles at the same speed as Anand.
He arrives at the park at the same time as Rahul.

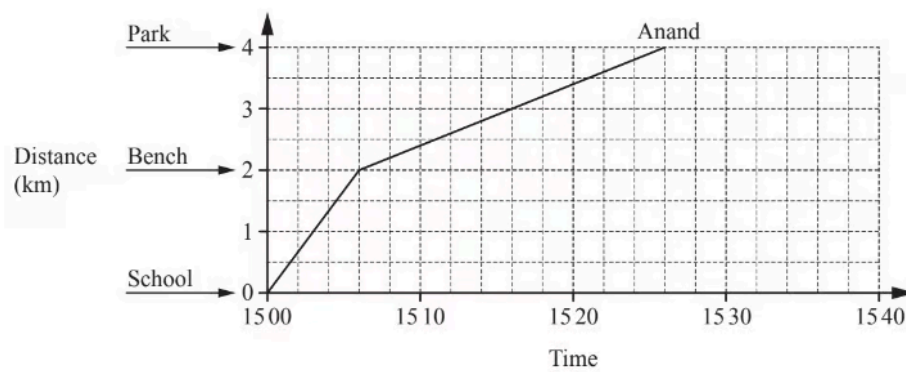
Find the time that Samir leaves school.

[1]

(4 marks)

(b) On another day, Anand cycles 2km to a bench and then walks the rest of the way to the park.

The travel graph shows his journey.



- i) Write down the ratio minutes cycling : minutes walking.
Give your answer in its simplest form.

[2]

- ii) Rahul leaves school at the same time as Anand.
Rahul walks 2 km to the bench at a constant speed of 6km/h.
He then cycles the rest of the way to the park at a constant speed of 15km/h.

- a) Complete the travel graph for Rahul's journey to the park.

[2]

- b) Use your travel graph to find the number of minutes that Anand arrives at the bench before Rahul.

..... min [1]

- c) Find the greatest distance between Anand and Rahul as they travel to the park.

..... km [1]

(6 marks)

15 (a) A cuboid measures 5cm by 4 cm by 2cm.

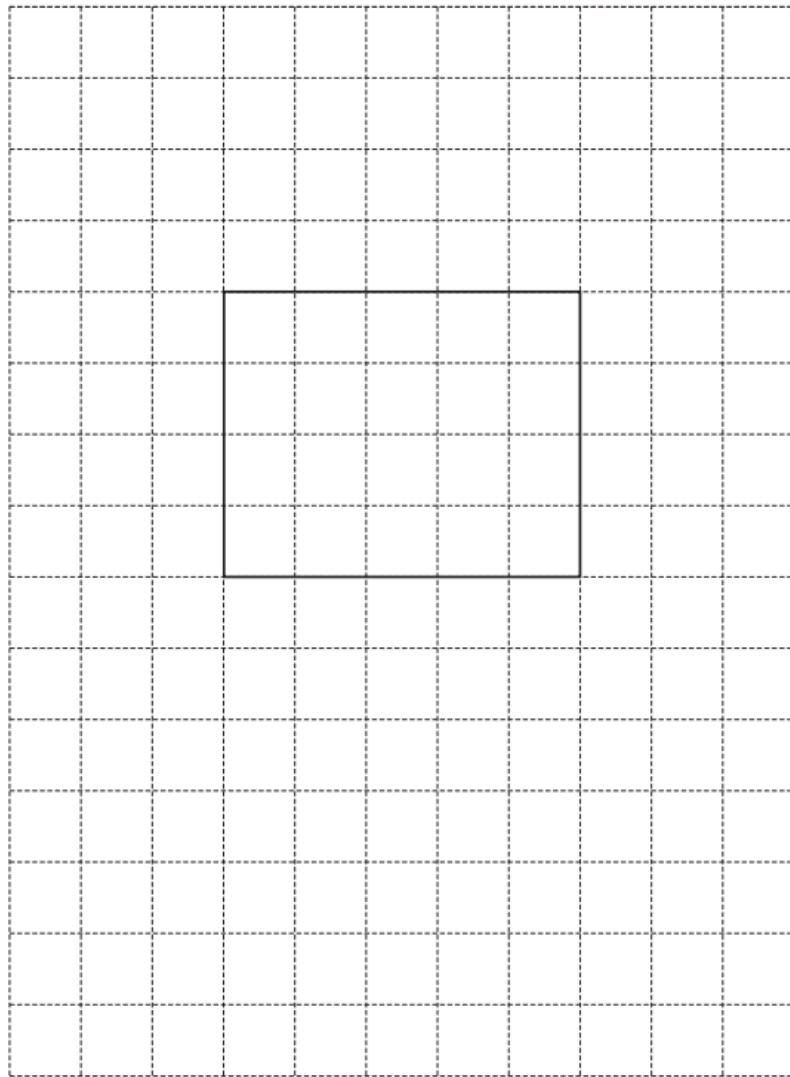
Calculate the volume of this cuboid.

Give the units of your answer.

(3 marks)

(b) On the 1 cm² grid, draw an accurate net of this cuboid.

One face has been drawn for you.



(3 marks)

- 16** A square has perimeter $12x$.

Find an expression, in terms of x , for the area of the square. Give your answer in its simplest form.

(3 marks)

- 17 A cone with height 14.8 cm has volume 275 cm^3 .

Calculate the radius of the cone.

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3} \pi r^2 h$.]

..... cm

(3 marks)

- 18 A farmer has 45 horses and 20 cows.

i) Write this as a ratio of horses : cows.
Give your answer in its simplest form.

[1]

ii) The farmer wants the ratio of horses : cows to equal 5 : 3.
He keeps his 45 horses and buys some more cows.

Work out the number of cows he must buy.

[2]

(3 marks)

19 Sonia wants to invest \$5000 for 6 years.

Bank A pays compound interest at a rate of 3.5% per year.

Bank B increases the \$5000 by 22% at the end of 6 years.

Which bank will give Sonia the most money at the end of 6 years and by how much? You must show all your working.

Bank A

Bank B

Bank will give \$ more money.

(5 marks)

20 Talan completes one lap of a track every 80 seconds.

i) Work out how many laps he can complete in one hour.

[2]

ii) Naima completes one lap of the same track every 88 seconds. Talan and Naima start running from point *A* on the track at the same time.
They each complete a number of laps of the track.

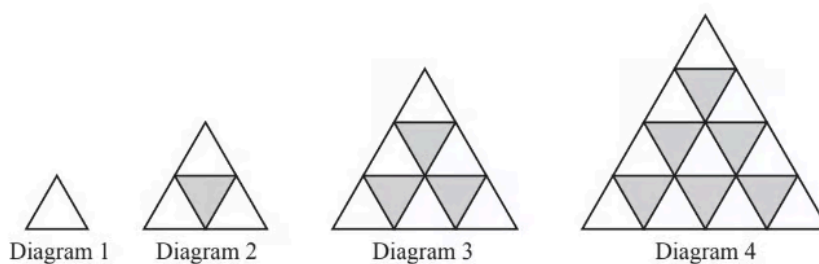
Work out the smallest number of laps they each complete before they are both at point *A* again at the same time.

Talan completes laps and Naima completes laps. [3]

(5 marks)

21 These are the first four diagrams in a sequence.

Each diagram is made from small equilateral triangles.



i) Write down the number of lines of symmetry of Diagram 3.

[1]

ii) Complete the table.

Diagram number (n)	1	2	3	4
Number of white triangles (w)	1	3	6	
Number of grey triangles (g)	0		3	
Total number of small triangles (t)	1	4		

[2]

iii) Find a formula, in terms of n , for the total number of small triangles, t , in Diagram n .

$t = \dots\dots\dots$ [1]

iv) The formula for the number of white triangles, w , in Diagram n is $w = \frac{1}{2}n(n+1)$.

Show that this formula gives the correct number of white triangles when $n = 3$.

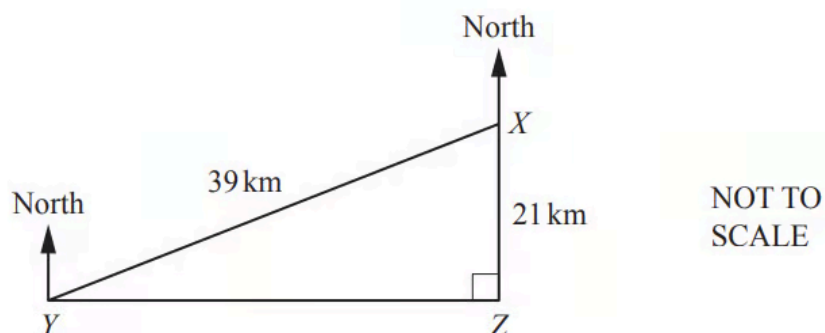
[2]

v) Complete this statement for Diagram 15.

When $n = 15$, $w = \dots\dots\dots$, $g = \dots\dots\dots$ and $t = \dots\dots\dots$ [3]

(9 marks)

22



A speedboat starts at X and travels to Y , then to Z and then back to X .

Z is due south of X and Y is due west of Z .

$XY = 39\text{ km}$ and $XZ = 21\text{ km}$.

i) Calculate YZ .

$YZ = \dots\dots\dots\text{km}$ [3]

ii) Calculate angle YXZ .

Angle $YXZ = \dots\dots\dots$ [2]

iii) Find the bearing of Y from X .

[1]

(6 marks)